

Curated Research Articles

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- **Mn-Mediated Reduction Coupling Mechanism Stabilizes Anion Redox Chemistry Toward High-Performance P2-Type Na-Layered Oxide Cathodes** — score: 1.000 Calcium doping in P2-type Na-layered oxide cathodes initiates a manganese-mediated reduction coupling mechanism that enhances anionic redox reactions through unique electron transfer from oxygen to manganese. This improvement leads to greater electrochemical performance, higher stability, and reduced lattice strain, enabling better capacity retention in sodium-ion batteries over extended cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Silicon-Based Anodes for Sulfide Solid-State Batteries: Failure Mechanisms and Multi-scale Design Strategies** — score: 1.000 The article reviews the challenges faced by silicon anodes in sulfide solid-state batteries, particularly the issues of electrochemical degradation and mechanical failure due to volumetric expansion. It proposes various engineering strategies at different levels to mitigate these problems, discusses the scalable production of sulfide solid electrolytes, and emphasizes the importance of in situ characterization techniques in understanding failure mechanisms and improving battery performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Molecular Engineering of Non-Covalent Networks in Multi-Functional Gel Electrolytes for High-Performance Solid-State Li O₂ Batteries** — score: 0.900 The article discusses the development of a multi-functional gel polymer electrolyte (PGFEN) for solid-state lithium-oxygen batteries, which enhances Li⁺ migration and redox kinetics through tailored non-covalent interactions. These interactions improve the electrolyte's performance by stabilizing the Li₂O₂ cycle and achieving a high ionic conductivity, reduced overpotential, and impressive discharge capacity, indicating significant potential for practical applications in battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Asymmetric Polarization Enabled Robust Framework of Prussian Blue Analog Cathode for Ultrastable Sodiation Lifespan Over 50 000 Cycles** — score: 0.800 This article presents a novel asymmetric polarization strategy to enhance the structural integrity of Prussian blue analogs used in sodium-ion batteries. By incorporating low-electronegative Mg²⁺, the framework is strengthened, resulting in an exceptional cycling lifespan of over 50,000 cycles and superior electrode performance, thereby addressing the common issue of framework degradation in battery materials. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Reliable Evaluation of Na-Ion Battery Materials: Eliminating Na-Metal Distortions** — score: 0.800 The article proposes utilizing NASICON-type counter electrodes instead of reactive sodium metal in Na-ion battery evaluations to eliminate distortions that compromise reproducibility and accuracy in key electrochemical indicators. This approach aims to standardize testing methods, facilitating more reliable material assessments and advancements in Na-ion battery development. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **LiMn_{0.5}Fe_{0.5}PO₄: A Cathode Material for Lithium-Ion Batteries: A Comprehensive Review** — score: 0.800 This review discusses LiMn_{0.5}Fe_{0.5}PO₄ as a promising cathode material for lithium-ion batteries, highlighting its balance of high voltage, energy density, and safety. It explores the structure-property relationships, phase behavior, and various strategies to overcome performance limitations, such as adjusting particle size, modifying surfaces, and designing composites, while calling attention to ongoing challenges and future research needs in enhancing its practical application. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Zn²⁺-Rich Microgel-Liquid Biphasic Electrolyte to Alleviate Interface Concentration Polarization for Advanced Aqueous Zinc Batteries** — score: 0.800 This study introduces a novel Zn²⁺-rich microgel-liquid biphasic electrolyte designed to alleviate concentration polarization at the zinc anode interface in aqueous zinc batteries. The microgel acts as a dynamic reservoir, replenishing Zn²⁺ to mitigate polarization and stabilize performance, while also suppressing polyiodide shuttling

when used with iodine-based full cells, leading to exceptional cycling stability and longevity in battery systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Dual characteristics of Mn^{2+} dissolution from manganese oxide cathode materials in aqueous zinc batteries: trade-offs between active material loss and double capacity contribution** — score: 0.800 The article discusses the dissolution behavior of Mn^{2+} ions in zinc-manganese oxide batteries, highlighting the trade-off between the loss of active material and the dual capacity contribution from dissolved Mn^{2+} . It aims to enhance the performance of aqueous batteries by systematically analyzing these interactions. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Interfacial strain engineering induces fatigue-resistant, high-capacity photo-assisted lithium storage** — score: 0.800 The article discusses how interfacial strain engineering in $\text{TiO}_2/\text{K}_2\text{Ti}_4\text{O}_9\text{-rGO}$ composites improves photo-assisted lithium storage capabilities. This method enhances charge separation and lithium ion transport while simultaneously reducing issues like cracking, lattice fatigue, and capacity fade, leading to more durable and efficient energy storage solutions. Journal: *RSC - EES Batteries latest articles*
- **Machine Learning Interatomic Potentials for Energy Materials: Architectures, Training Strategies, and Applications** — score: 0.800 The article reviews the development and applications of machine learning interatomic potentials (MLIPs) in the context of energy materials, focusing on their ability to combine quantum-level accuracy with efficiency in large-scale simulations. It discusses various MLIP architectures, training strategies, and their effectiveness across a range of energy systems, while also addressing challenges related to model validation and the influence of dataset choices on predictive accuracy. Emerging research directions, such as generative design and autonomous discovery, are highlighted as key areas for future exploration in energy materials research. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Understanding Trade-Off Between Zincophilicity and Hydrogen-Evolution: Toward Ultra-Stable Aqueous Zinc Metal Batteries** — score: 0.800 The article presents a novel approach to optimizing zinc metal anodes for aqueous batteries by balancing zincophilicity and hydrogen evolution reactions. Through the use of metal-organic framework precursors and pyrolysis, the study developed ZnBi@NCF alloys that demonstrate remarkable cycling stability and efficiency, achieving over 3600 cycles while effectively suppressing unwanted hydrogen evolution. This work advances the understanding of surface interactions in metal anodes, paving the way for more stable energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Unlocking hidden sodium** — score: 0.800 Recent research highlights the potential of Fe-based polyanionic cathodes for sodium-ion batteries, addressing the challenge of incomplete sodium utilization. By substituting vanadium in phosphate-based cathodes, the study reveals that this tuning enhances local sodium coordination, allowing for nearly total sodium usage, which improves energy density and cycling stability. Journal: *Nature Energy*
- **Harmonized sodium coordination engineering for high-energy phosphate cathodes** — score: 0.800 The article discusses the enhancement of sodium-ion batteries by optimizing the sodium coordination environment through targeted vanadium (V^{3+}) substitution in iron-based polyanionic cathodes. This approach activates previously inactive sodium sites and mitigates irreversible sodium loss, ultimately leading to improved performance and stability for high-voltage applications. Journal: *Nature Energy*
- **Multiscale Computational Insights for Hydrogel Electrolytes in Aqueous Zinc Metal Batteries** — score: 0.800 This review highlights the crucial impact of multiscale computational approaches in designing hydrogel electrolytes for aqueous zinc metal batteries. It explores the challenges these electrolytes face, such as stability and performance issues, while establishing structure-property relationships and interfacial dynamics critical for enhancing battery efficiency. The findings underscore the necessity for a comprehensive understanding of various scale interactions to advance the development of effective hydrogel electrolytes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A synergistic dual additive strategy for constructing gradient hydrophobic interfaces to-**

- ward ultrastable aqueous zinc metal anodes** — score: 0.800 The article presents a novel dual additive strategy that enhances the performance of aqueous zinc metal anodes by creating a multifunctional gradient hydrophobic interface, which effectively separates the roles of interfacial hydrophobicity and solvation coordination. This innovation significantly improves cycling stability, indicating promising advancements in battery technology. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Precursor-driven Jahn–Teller distortion as a hidden origin of surface instability in Mn-stabilized Ni-rich cathodes** — score: 0.800 The article investigates how the precursor-driven Jahn–Teller distortion in Mn-stabilized high-Ni cathodes leads to surface instability, primarily due to surface oxidation of Mn during the precursor stage. This newly identified interfacial destabilization mechanism highlights significant implications for the stability of battery materials. Journal: *RSC - Energy Environ. Sci. latest articles*
 - **Single-Atom Iron-Catalyzed Vapor-Phase Polymerization of Phosphorus Enables Uniform Nanoconfinement in Carbon Frameworks for Durable Li-Ion Batteries** — score: 0.800 The article discusses the use of single-atom iron catalysts to enhance the vapor-phase polymerization of red phosphorus for lithium-ion battery anodes. By achieving a uniform dispersion of phosphorus nanoparticles in a carbon framework, this method improves structural stability and prolongs capacity retention during charging cycles, addressing the challenges of low conductivity and volume expansion associated with red phosphorus. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **A Highly Hydrophilic Air Electrode With Water-Induced Surface Reconstruction for Efficient Reversible Protonic Ceramic Cell at Low Water Partial Pressures** — score: 0.800 This study introduces a novel air electrode made from BaCo_{0.45}Fe_{0.45}In_{0.1}O₃- for reversible protonic ceramic cells (R-PCCs), enabling efficient hydrogen production at low water vapor pressures. The electrode’s exceptional hydrophilicity facilitates significant water uptake and enhances electrochemical performance through in situ nanocatalyst exsolution. This advancement addresses challenges in hydrogen generation in water-scarce areas, showing high current densities and stable Faradaic efficiency across varying water conditions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Generating isomorphous intermediates overcomes synthesis barriers for scalable direct upcycling of LiFePO₄ battery cathodes** — score: 0.800 The article discusses a novel solid-state upcycling method that transforms used LiFePO₄ battery cathodes into high-performance LiM_{nx}Fe_{1-x}PO₄ by using isomorphous intermediates. This approach enhances material uniformity and performance while minimizing environmental impact and costs, paving the way for a sustainable, industrial-scale recycling process for lithium battery materials. Journal: *Joule*
 - **Advancing High-Performance Si Anodes for Next-Generation Li-Ion Batteries: Strategies for Structural Stability, Interfacial Compatibility, and Transport Kinetics** — score: 0.800 This review explores the advancements and challenges faced by silicon (Si) anodes in lithium-ion batteries, including volume expansion, interfacial instability, and low conductivity. It details modification strategies like nano-architectures and alloying to enhance performance and longevity, while also addressing industrialization hurdles and future research directions for integrating Si anodes into next-generation high-energy-density batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Truxenone-Based Covalent Organic Framework With Extended π -Conjugation and Multiple Active Sites for Symmetric All-COF Sodium-Ion Batteries** — score: 0.800 The article presents the development of a truxenone-based covalent organic framework (TRO-DH-COF) that features enhanced π -conjugation and multiple redox-active sites, which is combined with carbon nanotubes to create the first symmetric all-covalent organic framework sodium-ion batteries (ACSIBs). These batteries demonstrate impressive performance metrics, including high specific capacity, excellent rate capability, and long cycling durability, positioning this framework as a significant advancement in sodium-ion battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Competitive Coordination and Confinement in Rigid-Flexible Networks for High-Rate and Ultra-Stable Quasi-Solid-State Batteries** — score: 0.800 This research introduces a novel

quasi-solid electrolyte that effectively combines rigidity and flexibility, addressing challenges in high-voltage lithium-metal batteries. The unique polymer network enhances ion transport while suppressing dendrite formation, leading to significant improvements in cycling stability and energy density, demonstrated by impressive performance metrics such as over 4,000 hours of stable operation and sustained capacity through extensive cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Linking ion ordering, structural dynamics, and charge storage kinetics in the tunnel structure electrode material Li_xMoO_2** — score: 0.800 The article explores the relationship between ion ordering, the dynamic structural changes in Li_xMoO_2 , and the kinetics of charge storage in this tunnel-structured electrode material. It aims to enhance understanding of how these factors influence the efficiency and performance of energy storage systems, as reported in *Energy Storage Materials*. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Natural Anionic Interfacial Reconstruction for Suppressing Pitting Corrosion Enables Durable Seawater Zinc-Ion Batteries** — score: 0.700 The article discusses a newly developed molecular sieve-alginate interface layer that enhances the performance of zinc metal anodes in seawater-based electrolytes, effectively mitigating issues like pitting corrosion and dendrite growth. This innovative design enables zinc-ion batteries to achieve impressive long-term stability, with symmetric cells cycling over 2000 hours and full cells maintaining high capacity after extensive use, promising affordability and sustainability for large-scale energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Large-Pore Mesoporous Covalent Organic Frameworks Fast Ion Sieves by Kinetics-Mediated Micelle Assembly for Aqueous Batteries** — score: 0.700 The article discusses the development of large-pore mesoporous covalent organic frameworks (COFs) that function as effective ion sieves for aqueous batteries. These structures are designed to enhance the stability of Zn-I2 batteries by promoting uniform zinc ion deposition while preventing the interference of polyiodides, leveraging functional group interactions within the COFs. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Comprehensive Insight into the Synthesis, Structure, Stability, and Ionic Conductivity of $\text{Na}_5\text{MSi}_4\text{O}_{12}$ type Solid Electrolytes for Solid-State Sodium Batteries** — score: 0.700 The article reviews the synthesis, structural characteristics, stability, and ionic conductivity of $\text{Na}_5\text{MSi}_4\text{O}_{12}$ -type silicates, highlighting their potential as solid electrolytes for sodium batteries. These materials exhibit high ionic conductivity alongside strong chemical and electrochemical stability, making them promising candidates for enhancing battery performance. Journal: *RSC - EES Batteries latest articles*
- **Calendar and cycle ageing of commercial sodium-ion cells with layered oxide cathodes: how does operation influence degradation modes?** — score: 0.600 This article investigates the effects of operational conditions on the degradation modes of commercial sodium-ion batteries featuring layered oxide cathodes. The study highlights the potential of sodium-ion technology as an alternative to lithium-ion systems, examining how factors such as calendar and cycle ageing influence battery performance and longevity. Journal: *RSC - EES Batteries latest articles*
- **Breaking the Cascade Dissolution Loop through Self-inhibition Mechanism for Zinc Vanadium Oxide Battery** — score: 0.600 This article discusses a novel approach to improving zinc-vanadium oxide batteries by addressing the issue of cascade dissolution through a self-inhibition mechanism. The authors challenge previous claims regarding the benefits of pre-intercalating foreign species into the battery cathode and present evidence that enhances the stability of the V-O bond, significantly impacting battery performance. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Nanoarchitectonics for Green Upgrading Spent Battery With PFAS via One-Pot Mechanochemistry** — score: 0.600 The article presents a novel one-pot mechanochemical approach for upgrading spent lithium-ion battery materials using per- and polyfluoroalkyl substances (PFAS) through ball-milling. This method effectively cleaves C-F bonds in PFAS, transforms inactive FePO_4 into high-performance FeF_2 , and creates a conductive carbon layer, resulting in nanocomposites that exhibit exceptional battery capacity and cycling stability, thus promoting a circular economy by

turning waste into valuable resources. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Electrode surface engineering with electrolyte additives, improving reversibility of magnesium metal anode batteries** — score: 0.500 The article discusses advancements in rechargeable magnesium batteries (RMBs) by focusing on electrode surface engineering with electrolyte additives. This approach aims to enhance the reversibility of magnesium metal anodes, addressing the challenges posed by electrolyte limitations in the development of these high-capacity energy storage systems. Journal: *RSC - EES Batteries latest articles*
- **[ASAP] Lithium-Mediated Reconstruction Activates Carbon Additives into Ion-Electron Percolative Media for All-Solid-State Batteries** — score: 0.500 The article discusses a novel approach in all-solid-state batteries, employing lithium-mediated reconstruction to convert carbon additives into ion-electron percolative media. This method enhances the battery's performance by improving the conductivity and overall efficiency of the solid-state battery system. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **Li-argyrodite sulfide solid electrolytes: From fundamental research to industrialization** — score: 0.500 The article reviews recent developments in Li-argyrodite sulfide solid electrolytes for all-solid-state lithium batteries, addressing key challenges such as moisture sensitivity and high costs. It highlights progress in synthesis and interface stability and discusses the potential for industrial-scale production alongside existing barriers. Journal: *Joule*
- **[ASAP] Inducing Rapid Conversion of Sulfur Species Based on a Nanostructured Regulation Strategy for Li-S Batteries** — score: 0.500 The article discusses a novel strategy that utilizes nanostructured materials to accelerate the conversion processes of sulfur species in lithium-sulfur (Li-S) batteries. This approach aims to enhance the performance and efficiency of Li-S batteries, which are promising for energy storage due to their high energy density. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **Chemically Driven Conformational Rearrangement of PVDF-Based Polymer Electrolyte to improve ionic conductivity for Long-cycling Lithium Metal Batteries** — score: 0.400 The article explores a chemically driven conformational rearrangement method to enhance the ionic conductivity of PVDF-based solid polymer electrolytes (SPEs) for long-cycling lithium metal batteries. Despite their notable thermal stability and film-forming capabilities, PVDF-based SPEs typically suffer from low ionic conductivity and side reactions at interfaces, which this research aims to address. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Secondary alkali-metal salts in supporting electrolytes for lithium polysulfide flow batteries** — score: 0.400 This article explores the use of secondary alkali-metal salts in the electrolytes of lithium polysulfide flow batteries, highlighting their potential to enhance the performance and efficiency of these next-generation energy storage systems. The study presents insights into how these salts can improve the stability and electrochemical behavior of lithium polysulfide (LiPS) solutions, positioning them as a viable alternative to traditional lithium-sulfur battery configurations. Journal: *RSC - EES Batteries latest articles*
- **Scaling up Si-C composite synthesis from recycled graphite for high-energy-density and low-environmental-impact batteries** — score: 0.400 The article discusses the scalable synthesis of graphite-silicon (GtSi) composites from recycled graphite, aimed at enhancing the performance of lithium battery anodes by achieving a high specific capacity of around 1000 mAh/g. This approach addresses both energy density needs and environmental considerations in battery production. Journal: *RSC - EES Batteries latest articles*
- **Disentangling the Multiscale Degradation Pathways in Fe-N-C Fuel Cells With Advanced Characterization Techniques** — score: 0.400 The article reviews the complex degradation mechanisms affecting Fe-N-C catalysts in proton exchange membrane fuel cells, which are considered promising alternatives to platinum-based catalysts. By utilizing advanced characterization techniques, the review presents a framework for understanding the multiscale degradation processes and stresses the importance of this knowledge in developing stabilization strategies, ultimately aiming to enhance the

durability and commercialization potential of Pt-free fuel cells. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **A gas-solid hydride ion battery and a hydrogen-electricity co-storage system** — score: 0.400 The article presents a novel gas-solid hydride ion battery that enhances the thermochemical hydrogen storage process by transitioning it into an electrochemical format, resulting in improved energy efficiency. This innovative battery decouples reactions, allowing for high storage capacities and stable cycling performance, making it a promising solution for hydrogen-electricity co-storage systems. Journal: *Joule*
- **Neutral-cluster-mediated suppression of solvent migration in high-voltage ether-based electrolytes** — score: 0.400 The article discusses the role of neutral clusters in mitigating solvent migration within high-voltage ether-based electrolytes, which is significant for enhancing the performance and stability of energy storage systems. The authors investigate the mechanisms by which these clusters operate, potentially leading to improved efficiency in electrochemical applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] Alternating Current–Electrophoretic Deposition-Assembled Metal–Organic Framework-Derived LiFePO₄ Composite Films: Robust Redox Kinetics in Binder- and Carbon Black-Free Electrodes** — score: 0.400 This article discusses the development of LiFePO₄ composite films using alternating current-electrophoretic deposition of metal-organic frameworks, resulting in electrodes that are free of binders and carbon black. The study highlights the enhanced redox kinetics of these films, suggesting their potential for use in high-performance energy storage applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Robust Solid-Electrolyte Interphase via Si–O Molecular Engineering for Durable Rechargeable Magnesium Batteries** — score: 0.400 The article discusses advancements in rechargeable magnesium batteries, focusing on the development of a robust solid-electrolyte interphase (SEI) achieved through Si–O molecular engineering. This innovation aims to enhance the durability and performance of the batteries, addressing common issues related to electrolyte stability and longevity. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **[ASAP] Ultrahigh-Rate All-Solid-State Batteries with a Dendrite-Free LiAl Anode via Compositional and Structural Engineering** — score: 0.400 The article presents the development of ultrahigh-rate all-solid-state batteries featuring a dendrite-free lithium aluminum (LiAl) anode. Through innovative compositional and structural engineering, the researchers achieved enhanced battery performance, potentially advancing the technology for practical energy storage applications. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **[ASAP] Carbon-Doped Co₃O₄ Derived from Coordination Polymers as Catalysts for the Oxygen Electrode in Metal–Air Batteries: ORR Activation through the OER** — score: 0.400 The article presents the development of carbon-doped Co₃O₄ catalysts derived from coordination polymers, aimed at enhancing the performance of the oxygen electrode in metal-air batteries. It highlights the catalysts' ability to activate the oxygen reduction reaction (ORR) through improved oxygen evolution reaction (OER) processes, showing potential for increased efficiency in energy storage applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] High-Performance Double-Atom Catalysts with Accelerated Na₂S Oxidation and Sulfur Reduction for Improved Cathode Performance in Na–S Batteries** — score: 0.400 The article discusses the development of high-performance double-atom catalysts that enhance the oxidation of Na₂S and sulfur reduction, leading to significant improvements in the cathode performance of sodium-sulfur (Na–S) batteries. This advancement addresses key challenges in energy storage technology by optimizing electrochemical reactions. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Bifunctional Bimetallic Biomass-Derived Electrocatalyst Materials for Usage as Air Electrode Catalysts in Rechargeable Zinc–Air Batteries** — score: 0.300 The article presents a study on bifunctional bimetallic electrocatalysts derived from biomass, which are evalu-

ated for their performance as air electrode catalysts in rechargeable zinc–air batteries. The findings indicate their potential to enhance the efficiency and sustainability of energy storage systems. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*

- **Toward Stable Aqueous Zinc Batteries: Reconstruction of the Coordination Environment of Electrolytes by the addition of Cationic Polymers with Multiple Coordination Modes** — score: 0.300 The article discusses the development of stable aqueous zinc batteries by enhancing the coordination environment of electrolytes through the incorporation of cationic polymers that possess multiple coordination modes. This approach aims to improve the performance and longevity of zinc batteries, addressing challenges associated with their stability. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] Upcycling Waste Polyethylene into Defect-Engineered CNTs via Pyrolysis-CVD for Ultrastable Flowless Zinc–Bromine Batteries** — score: 0.300 The article discusses a novel process for transforming waste polyethylene into defect-engineered carbon nanotubes (CNTs) using pyrolysis followed by chemical vapor deposition (CVD). These CNTs are then applied in ultrastable flowless zinc-bromine batteries, potentially advancing energy storage technology by improving battery efficiency and sustainability. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **[ASAP] High-Entropy Metal Single-Atom Catalysts Anchored in Hierarchically Porous Carbon for Accelerated Sulfur Conversion in Lithium–Sulfur Batteries** — score: 0.300 This study presents the development of high-entropy metal single-atom catalysts integrated into hierarchically porous carbon structures, aimed at enhancing the efficiency of sulfur conversion in lithium-sulfur batteries. The innovative approach seeks to improve battery performance through advanced catalytic materials, addressing challenges in energy storage technology. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **[ASAP] Synergistic Regulation of Electrolyte Composition and Electrode Interface for Enhanced Low-Temperature Performance of Lithium-Ion Batteries** — score: 0.300 The article discusses a novel approach to improve the low-temperature performance of lithium-ion batteries by optimizing the electrolyte composition and electrode interface. It highlights the synergistic effects of these adjustments, which lead to enhanced battery efficiency at reduced temperatures, potentially expanding the operational range and lifespan of these energy storage systems. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Substitutional Design of Nanosized -MnO₂ Cathode for Rechargeable Magnesium Battery** — score: 0.300 The article presents a novel approach to designing nanosized -MnO₂ cathodes for rechargeable magnesium batteries through substitutional strategies. It highlights the impact of structural modifications on the electrochemical performance, demonstrating significant improvements in capacity and cycle stability, thereby advancing the development of more efficient energy storage solutions. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*